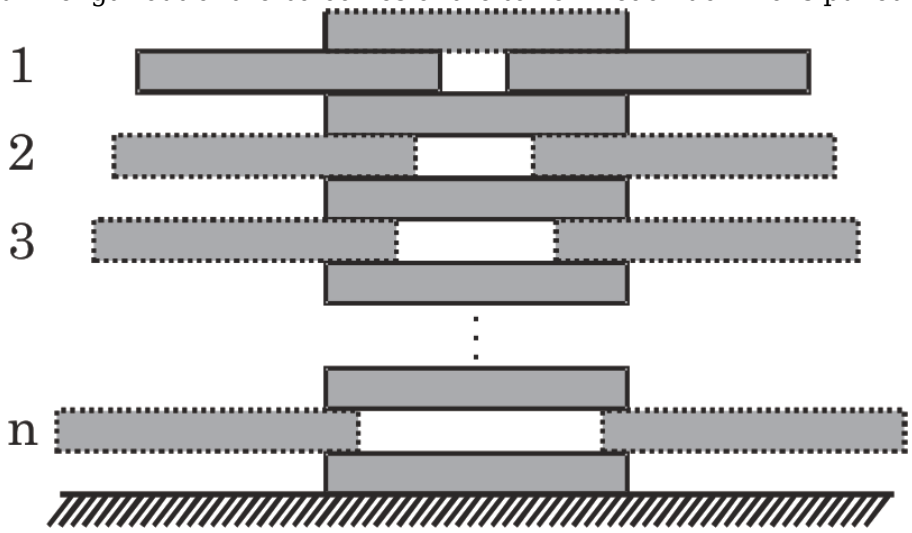
PART-1 : PHYSICS

SECTION-I

1) The n-storey symmetrical tower shown in the figure was built from dominoes of length □. What is the maximum length out of the balconies of the tower if each domino is pulled out as much as possible ?



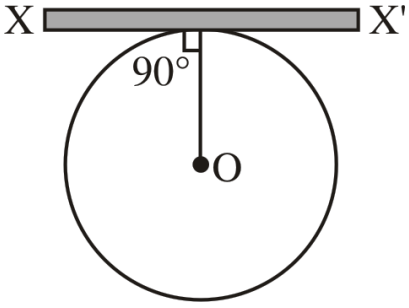
(A) (3n – 1)□/3n

(B) (2n – 1)□/2n

(C) (n – 1)□/n

(D) 5□/8 for each balcony

2) A thin wire of length L and uniform linear mass density ρ is bent into a circular loop with centre at O as shown. The moment of inertia of the loop about the axis XX' (lies in the plane of loop) is :-



(A) $\frac{\rho L^3}{8\pi^2}$

(B) $\frac{\rho L^3}{16\pi^2}$

(C) $\frac{5\rho L^3}{16\pi^2}$

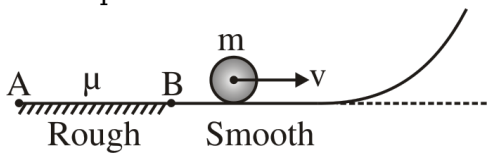
(D) $\frac{3\rho L^3}{8\pi^2}$

For More Material Join: @JEEAdvanced_2025

3) ***Assertion*** **:** The total distance moved by any point on the periphery of a wheel of radius R along the surface in one revolution is 2πR.
Reason **:** In rolling motion of a wheel, every point on its periphery comes in contact with the surface once in one revolution.

(A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(B) If both Assertion & Reason are True but Reason is not a correct explanatilon of the Assertion.
(C) If Assertion is True but the Reason is False.
(D) If both Assertion & Reason are false.

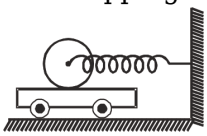
4) A disc of mass m is rolling without slipping on a smooth horizontal surface followed by a smooth curved track. It reaches a maximum height on the ramp and then returns on the horizontal track.



Friction force acting on the disc in part BA is :-

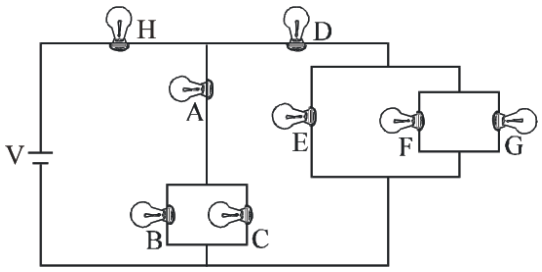
(A) μmg
(B) $\frac{2}{3}\mu mg$
(C) $\frac{3}{2}\mu mg$
(D) zero

5) In the figure shown, the mass of the disc as well as that of the trolley is M. The spring is ideal and has stiffness k. The trolley can move horizontally on the floor without friction, and the disc can roll on the trolley surface without slipping. The spring is compressed and the system released so that oscillations begin. The :-



(A) Acceleration of center of disc = twice of that of trolley
(B) Acceleration of center of disc = thrice of that of trolley
(C) Acceleration of center of disc = half of that of trolley
(D) Acceleration of center of disc = that of trolley

6) Circuit shows eight identical bulbs (A to H) connected through a battery of emf V. If bulb H is replaced by a wire with no resistance then rank the brightness of the bulbs, from brightest to dimmest.



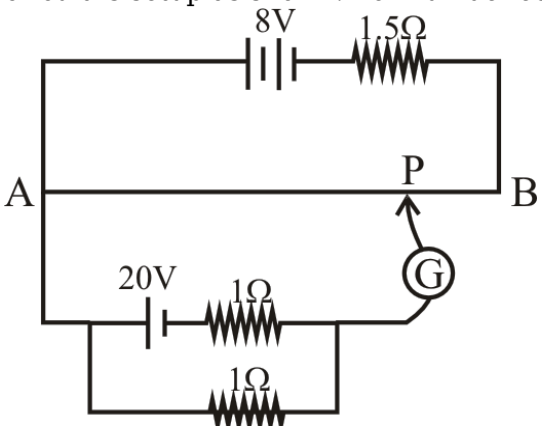
(A) A > D > E = F = G > B = C
(B) D > A > B = C > E = F = G

For More Material Join: @JEEAdvanced_2025

(C) D > A > E = F = G > B = C

(D) A < D < E = F = G < B = C

7) A potentiometer has a resistance wire of 10 m length and resistance 10Ω as shown in figure. A circuit is setup as shown. For null deflection in galvanometer length AP is :



(A) 2m

(B) 4.5 m

(C) 6.5 m

(D) None of these

8) A 15Ω resistance is connected in the left gap and an unknown resistance less than 15Ω is connected in the right gap of a meter bridge. When the resistances are interchanged, the balance point is found to shift by 20 cm. The unknown resistance is

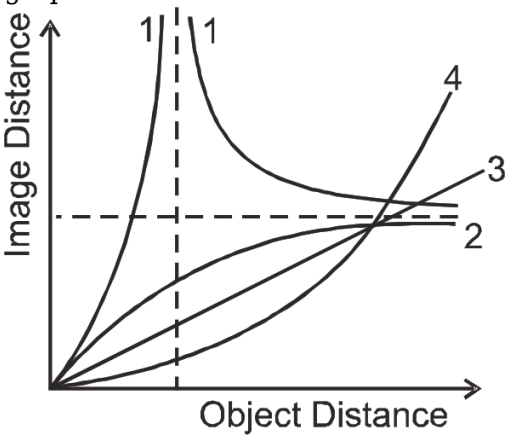
(A) 5 Ω

(B) 6 Ω

(C) 8 Ω

(D) 10 Ω

9) A small particle is placed at the pole of a concave mirror and then moved along the principal axis to a large distance. During the motion, the distance between the pole of the mirror and the image is measured. The procedure is then repeated with a convex mirror, a concave lens and a convex lens. The graph is plotted between image distance versus object distance. Match the curves shown in the graph with the mirror or lens that is corresponding to it. (Curve 1 has two segments)



List - I	List - II

For More Material Join: @JEEAdvanced_2025

(P)	Converging lens	(1)	1
(Q)	Converging Mirror	(2)	2
(R)	Diverging Lens	(3)	3
(S)	Diverging Mirror	(4)	4

(A) P → 1; Q → 1; R → 2; S → 2

(B) P → 1; Q → 2; R → 3; S → 4

(C) P → 1; Q → 2; R → 2; S → 3

(D) P → 3; Q → 2; R → 1; S → 4

10) A galvanometer has coil of resistance 100Ω showing a full scale deflection 50 mA. What resistance should be added to use it as ammeter of range 10A?

(A) $\frac{100}{199}$

(B) $\frac{199}{100}$

(C) $\frac{10}{9}$

(D) $\frac{9}{10}$

11) The length of a wire increased by 2% by stretching. Then the percentage change in its resistance will be :-

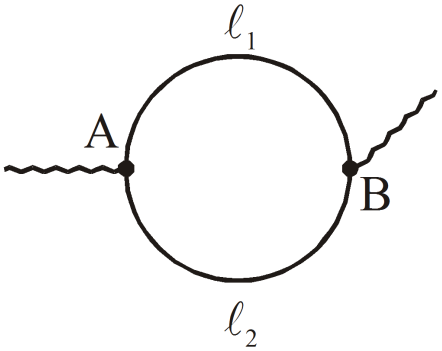
(A) 2%

(B) 4%

(C) $\frac{1}{2}$%

(D) $\frac{1}{4}$%

12) A ring is made of a wire having a resistance $R_0 = 15\ \Omega$. Find the points A and B as shown in figure at which a current carrying conductor should be connected so that the resistance R of the sub circuit between these point is equal to $\frac{10}{3}\ \Omega$.



(A) $\frac{\ell_1}{\ell_2} = \frac{1}{2}$

For More Material Join: @JEEAdvanced_2025

(B) $\frac{\ell_1}{\ell_2} = \frac{6}{5}$

(C) $\frac{\ell_1}{\ell_2} = \frac{4}{5}$

(D) $\frac{\ell_1}{\ell_2} = \frac{1}{4}$

13) The focal lengths of objective lens and eye lens of a Galilean telescope are respectively 30 cm and 3.0 cm. Telescope produces virtual, erect image of an object situated far away from it at least distance of distinct vision from the eye lens. In this condition, the magnifying power of the Galilean telescope should be :-

(A) +11.2

(B) –11.2

(C) –8.8

(D) +8.8

14) A compound microscope has a magnification of 30. The focal length of the eyepiece is 5 cm. If the final image is formed at the least distance of vision (25 cm), the magnification produced by the objective is

(A) 5

(B) 7.5

(C) 10

(D) 15

15) **Statement-1:** Voltage across a battery is always less than its emf.
Statement-2 : There is always some loss of energy due to internal resistance of the battery.

(A) Statement–1 is True, Statement–2 is True ; Statement–2 is a correct explanation for Statement–1.

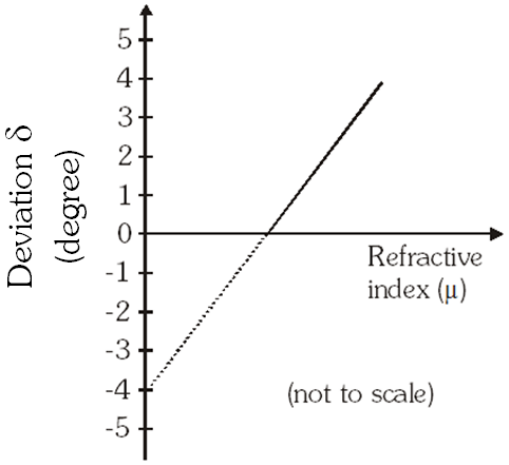
(B) Statement–1 is True, Statement–2 is True ; Statement–2 is not a correct explanation for Statement–1.

(C) Statement–1 is True, Statement–2 is False.

(D) Statement–1 is False, Statement–2 is True.

16) Figures shows the graph of angle of deviation δ versus refractive index μ of the material of constant thin angled prisms corresponding to light rays incident at a small angle of incidence. The

For More Material Join: @JEEAdvanced_2025



prism angle and slope of the line are respectively :

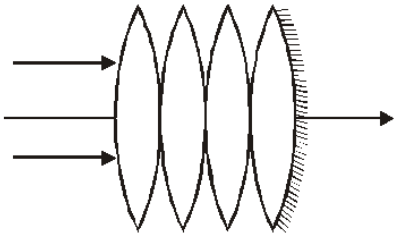
(A) 4^0 and 2

(B) 2^0 and 1/2

(C) 2^0 and 4

(D) 4^0 and 4

17) Four identical lenses are kept one beside the other on the same optical axis as shown in the figure. The right surface of rightmost lens is silvered. Focal length of each lens is 20 cm and radius of the silvered surface is 20 cm. The focal length of the combined system is



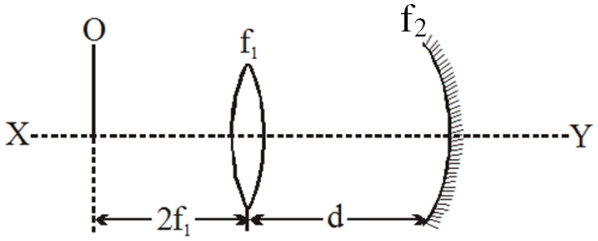
(A) 2 cm

(B) –2 cm

(C) 5 cm

(D) –5 cm

18) An erect object O is placed transverse to the axis of a converging lens at a distance equal to twice the focal length f_1 of the lens. On the other side of the lens, is placed a converging mirror of focal length f_2 at a distance d from the lens. The lens and the mirror have a common optic axis XY. It is observed that the image of the object O seen after one refraction from lens and one reflection from mirror is erect, real and of the same size as the object O. Then the distance d is given by :-



(A) $d = f_1 + f_2$

(B) $d = 2 (f_1 + f_2)$

For More Material Join: @JEEAdvanced_2025

(C) $d = \frac{f_1 + f_2}{2}$

(D) $d = 2f_2$

19) The vernier of a vernier scale is divided into 10 divisions which coincide with 9 divisions of the main scale, each main scale division being 0.5 mm. When the two jaws of the instrument are in contact with each other, the 4th division of the vernier scale coincides with a main scale division and the zero of the vernier lies to the right of the zero of the main scale. When a sphere is inserted between the jaws, the zero of vernier scale lies slightly to the right of 1.8 cm and the sixth vernier division coincides with a main division. The diameter of sphere will be

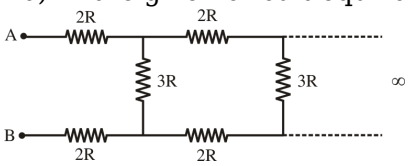
(A) 1.850 cm

(B) 1.810 cm

(C) 1.750 cm

(D) 1.710 cm

20) In the given circuit equivalent resistance between A and B is nR. Find the value of n.



(A) 6

(B) 7

(C) 8

(D) 9

SECTION-II

1) A vernier calipers has 1 mm marks on the main scale. It has 10 equal divisions on the vernier scale which match with 8 main scale divisions. If the least count of vernier calipers is $x \times 10^{-1}$ mm, find x.

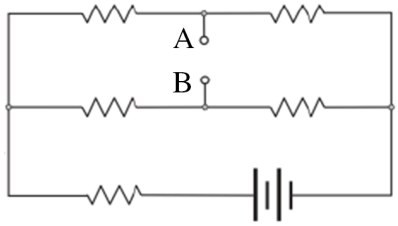
2) In an experiment of measuring the refractive index of a glass slab using travelling microscope in physics lab, a student measures real thickness of the glass slab as 5.25 mm and apparent thickness of the glass slab at 5.00 mm. Travelling microscope has 20 divisions in one cm on main scale and 50 divisions on Vernier scale is equal to 49 divisions on main scale. The estimated uncertainty in the measurement of refractive index of the slab is $\frac{x}{10} \times 10^{-3}$, where x is _____

3)

In a Searle's experiment, the diameter of the wire as measured by a screw gauge of least count 0.001 *cm* is 0.050 *cm*. The length, measured by a scale of least count 0.1 *cm*, is 110.0 *cm*. When a weight of 50 *N* is suspended from the wire, the extension is measured to be 0.125 *cm* by a micrometer of least count 0.001 *cm*. Maximum error in the measurement of Young's modulus of the material of the wire from these data is $P \times 10^9$ N/m^2, then value of *P* is

For More Material Join: @JEEAdvanced_2025

4) In the shown figure contains five unknown resistances and a battery. When ammeter is connected between the terminals A and B its reading is 6A and when a resistance of $4.5\ \Omega$ connected in series with the ammeter its reading becomes 3A. Now the ammeter and the resistance are disconnected and an ideal voltmeter is connected between terminal A and B. What would the voltmeter reading in volts?



5) The refractive indices of flint glass for red and violet light are 1.613 and 1.632 respectively. Find the value of (1000) × (θ) (in degree), where θ is angular dispersion produced by a thin prism of flint glass having refracting angle 5 degree.

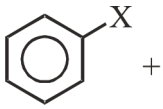
PART-2 : CHEMISTRY

SECTION-I

1) **Assertion :** Nucleophilic substitution reaction on an optically active halide gives a mixture of enantiomers.
Reason : The reaction should be in accordance with S_N1 mechanism.

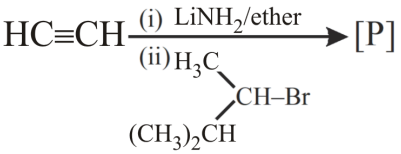
(A) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(B) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(C) Assertion is true but Reason is false.
(D) Both Assertion and Reason are false.

2)

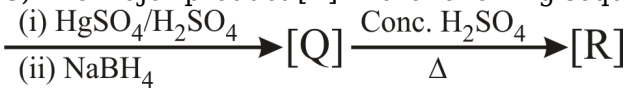
(A)	R–OH + $SOCl_2$	(I)	Swarts reaction
(B)	R–X+NaI in acetone (X=Cl, Br)	(II)	Darzen process
(C)	R–Br + AgF	(III)	Wurtz fittig
(D)	 + Na	(IV)	Finkelstein reaction

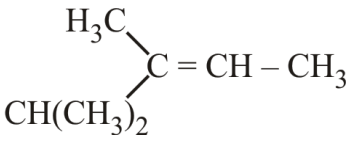
Correctly matched option is :
(A) (A) → (IV)
(B) (B) → (II)
(C) (D) → (III)
(D) (C) → (I)

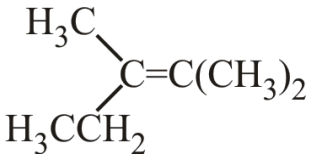
For More Material Join: @JEEAdvanced_2025

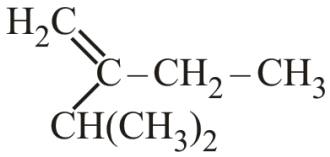


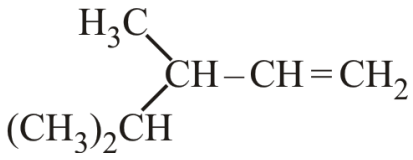
3) The major product [R] in the following sequence of reactions is :-

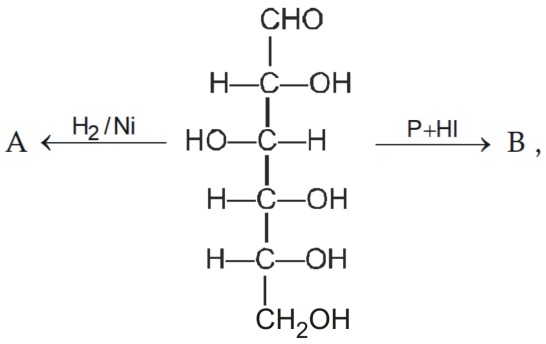


(A) 

(B) 

(C) 

(D) 

4) 

A and B can be :

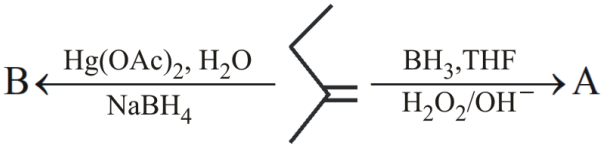
(A) Both are n-Hexane

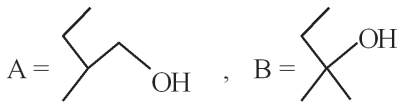
(B) Both are D-sorbitol

(C) A is n-Hexane B is D-sorbitol

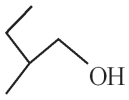
(D) A is D-sorbitol and B is n-Hexane

5) Find out the major products from the following reactions.

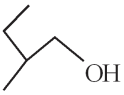
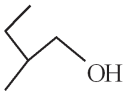


(A) 

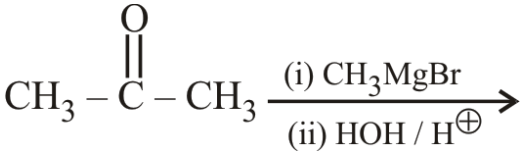
For More Material Join: @JEEAdvanced_2025

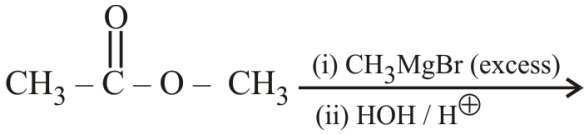
(B) A =  , B = 

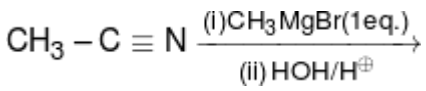
(C) A =  , B = 

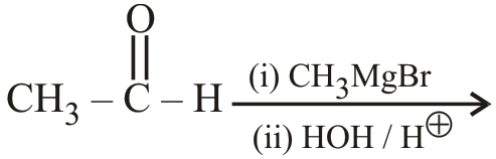
(D) A =  , B = 

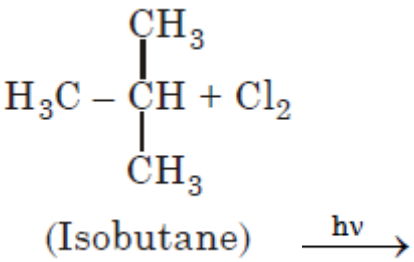
6) Which of the following gives 2° alcohol as major product :

(A) 

(B) 

(C) 

(D) 

7)  products

Which of the following statements is correct?

(A) Tertiary butyl chloride is formed as major product

(B) This reaction is example of nucleophilic substitution (S_N1)

(C) Primary alkyl chloride is formed as major product

(D) More than two alkyl chloride formed as final products

8)

Which of the following has been arranged in order of decreasing freezing point :-
(Assume α = 1 and molarity = molality)

(A) 0.05 M KNO_3 > 0.04 M $BaCl_2$ > 0.140 M sucrose > 0.075 M $CuSO_4$

(B) 0.04 M $BaCl_2$ > 0.140 M sucrose > 0.075 M $CuSO_4$ > 0.05 M KNO_3

(C) 0.075 M $CuSO_4$ > 0.140 M sucrose > 0.04 M $BaCl_2$ > 0.05 M KNO_3

For More Material Join: @JEEAdvanced_2025

(D) 0.075 M $CuSO_4$ > 0.05 M KNO_3 > 0.140 M sucrose > 0.04 M $BaCl_2$

9) **Assertion :** Equilibrium constant for endothermic reaction increase with increase in temperature.
Reason : Rate constant increases with increase in temperature.

(A) Both assertion and reason are true and the reason is a correct explanation of assertion.
(B) Both assertion and reason are true but reason is not a correct explanation of assertion.
(C) Assertion is true but the reason is false.
(D) Assertion is false but the reason is true.

10) In the system, $LaCl_3(s) + H_2O(g) + \text{heat} \rightleftharpoons LaClO(s) + 2HCl(g)$, equilibrium is established. More water vapour is added to disturb the equilibrium. If the pressure of water vapour at new equilibrium is double of that at initial equilibrium, the factor by which pressure of HCl is changed is:

(A) 2 times
(B) $\sqrt{2}$ times
(C) $\frac{1}{\sqrt{2}}$ times
(D) 4 times

11) A solution contains $[Cl^-] = [Br^-] = [I^-] = 10^{-2}$ M (initially). Now to this solution, $[AgNO_3]$ is added gradually.
Calculate concentration of Ag^+ ion when first salt starts to precipitate out of solution ?
Given:
(K_{sp} of [AgI] = 10^{-17}, K_{SP} of [AgBr] = 10^{-13},
K_{SP} of [AgCl] = 10^{-10})

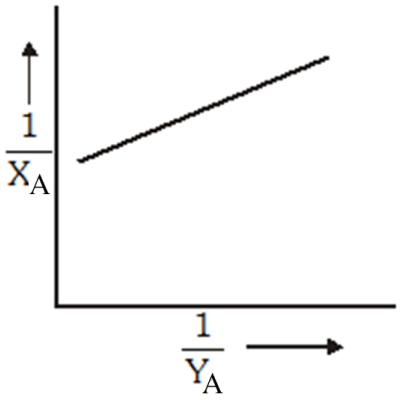
(A) 10^{-8}
(B) 10^{-10}
(C) 10^{-15}
(D) 10^{-7}

12) If first dissociation of $X(OH)_3$ is 100% whereas second dissociation is 50% and third dissociation is negligible then the pH of 4×10^{-3} M $X(OH)_3$ is :
(log2 = 0.30, log3 = 0.48)

(A) 11.78
(B) 10.78
(C) 2.5
(D) 2.22

13) Consider two liquids A and B having pure vapour pressure p_A^o and p_B^o forming an ideal solution. The plot of $\frac{1}{X_A}$ (Y-axis) vs $\frac{1}{Y_A}$ (X-axis) (where X_A and Y_A are the mole fraction of liquid A in liquid and vapour phase respectively) is linear with slope and intercept $\left(\text{on } \frac{1}{X_A}\text{axis}\right)$ respectively.

For More Material Join: @JEEAdvanced_2025



(A) $\frac{p_A^\circ}{p_B^\circ}$ and $\frac{p_A^\circ - p_B^\circ}{p_B^\circ}$

(B) $\frac{p_A^\circ}{p_B^\circ}$ and $\frac{(p_B^\circ - p_A^\circ)}{p_B^\circ}$

(C) $\frac{p_B^\circ}{p_A^\circ}$ and $\frac{p_A^\circ - p_B^\circ}{p_B^\circ}$

(D) $\frac{p_B^\circ}{p_A^\circ}$ and $\frac{p_B^\circ - p_A^\circ}{p_B^\circ}$

14) Select the **CORRECT** reactant for brown ring test of NO_3^- ion

(A) Aqueous solution of salt + ferrous sulphate solution + dil. H_2SO_4

(B) Aqueous solution of salt + ferric sulphate solution + conc. H_2SO_4

(C) Aqueous solution of salt + ferrous sulphate solution + conc. H_2SO_4

(D) Aqueous solution of salt + ferric sulphate solution + dil. H_2SO_4

15) Which halide does **NOT** produce halogen when heated with MnO_2 in dil. H_2SO_4 :-

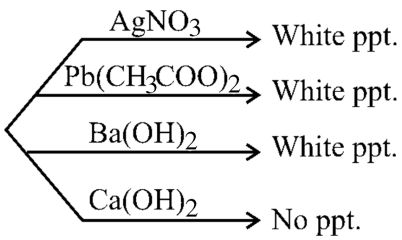
(A) F^-

(B) Cl^-

(C) Br^-

(D) I^-

16) Anion in Aq. solution



anion in Aqueous solution is -

(A) CO_3^{2-}

(B) SO_3^{2-}

For More Material Join: @JEEAdvanced_2025

(C) Cl^-

(D) $S_2O_3^{2-}$

17) Na_2CO_3 and $NaHCO_3$ salts are not distinguished by :-

(A) $MgCl_2$

(B) Hph

(C) $HgCl_2$(aq.)

(D) HCl

18) Which of the following cation does not produce red shaded flame colour in the flame test?

(A) Li^+

(B) Cu^{2+}

(C) Ca^{2+}

(D) Sr^{2+}

19) In an alkaline solution, sodium nitroprusside gives a violet colour with :

(A) SO_4^{2-}

(B) S^{2-}

(C) NO_2^-

(D) $S_2O_3^{2-}$

20)

List-I	List-II
(P) $Zn[Fe(CN)_5(NOS)]$	(1) Gives brown colour ppt with NH_3
(Q) $K_2[HgI_4]$ + KOH	(2) Gives pungent odour with Devarda Alloy in presence of alkaline medium
(R) $NaHC_4H_4O_6$	(3) Gives unknown compound of red colour with SO_2
(S) NO_3^-	(4) Gives white ppt. with K^+

(A) P → 3 ; Q → 2 ; R → 4 ; S → 1

(B) P → 2 ; Q → 1 ; R → 3 ; S → 4

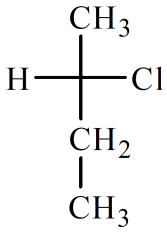
(C) P → 3 ; Q → 1 ; R → 2 ; S → 4

(D) P → 3 ; Q → 1 ; R → 4 ; S → 2

SECTION-II

1) How many total dichloro products of following structure on chlorination (including stereoisomerism).

For More Material Join: @JEEAdvanced_2025



2) A gaseous mixture of vapours of A and B, under a total pressure of 0.8 atm is in equilibrium with an ideal liquid solution of A(□) and B(□) . The mole fraction of A is 0.5 in the vapour phase and 0.2 in the liquid phase. The vapour pressure of pure liquid A is (in atm) _____ . (Nearest integer)
(Assume vapours behave ideally)

3) The equilibrium constant K_c at 298 K for the reaction A + B ⇌ C + D is 100. Starting with an equimolar solution with concentrations of A, B, C and D all equal to 1M, the equilibrium concentration of D is____ $\times 10^{-2}$ M. (Nearest integer)

4) Two salts A_2X and MX have the same value of solubility product of 4.0×10^{-12}. The ratio of their molar solubilities i.e. $\frac{S(A_2X)}{S(MX)}$ = _____. (Round off to the Nearest Integer).

5) How many of the following precipitates are black in colour ?
PbS, CuS, Bi_2S_3, $Fe(OH)_3$, CoS, NiS, Sb_2S_5, $Cr(OH)_3$, MnS, $BaCO_3$, $SrCO_3$

PART-3 : MATHEMATICS

SECTION-I

1) Consider parallelogram constructed on the vectors $\vec{P} = 5\vec{a} + 2\vec{b}$ and $\vec{Q} = \vec{a} - 3\vec{b}$. If $|\vec{a}| = 2\sqrt{2}$, $|\vec{b}| = 3$, $\vec{a} \wedge \vec{b} = \frac{\pi}{4}$, then length of smallest diagonal of the parallelogram is

(A) $\sqrt{261}$
(B) 13
(C) $\sqrt{216}$
(D) 15

2) Let $f(x) = (|x| - 1)\text{sgn}(x-1)$, then $f(x)$ is

(A) discontinuous at x = 0 but continuous at x = 1
(B) continuous at x = 0 but discontinuous at x = 1
(C) continuous at x = 0,1
(D) discontinuous at x = 0,1

3) Consider the function $f(x)$ such that $f(x-1) + f(x+3) = f(x+1)$ and $f(5) = 1$.

For More Material Join: @JEEAdvanced_2025

Assertion (A) : $\sum_{r=0}^{90} f(5+12r) = 90$.
Reason (R): $f(x)$ is a periodic function with period 12.

(A) **A** is correct but **R** is not correct.
(B) **A** is not correct but **R** is correct.
(C) Both **A** and **R** are correct but **R** is **NOT** the correct explanation of **A.**
(D) Both **A** and **R** are correct and **R** is the correct explanation of **A.**

4) The vector $\vec{P}$ is perpendicular to the vectors $\vec{A} = \hat{i} + 2\hat{j} + 3\hat{k}$ and $\vec{B} = -2\hat{i} + 4\hat{j} + \hat{k}$ and satisfies the condition $\vec{P}.\left(\hat{i} - 2\hat{j} + \hat{k}\right) = -6$, then $\left(\vec{P}.\hat{i} + \vec{P}.\vec{k}\right)$ is equal to

(A) 1
(B) 0
(C) –1
(D) 9

5) If n(U) = 600, n(A) = 100, n(B) = 200 and n(A ∩ B) = 50, then $n(\bar{A} \cap \bar{B})$ is -
(U is universal set and A and B are subsets of U)

(A) 300
(B) 350
(C) 250
(D) 200

6) Let $\vec{p}, \vec{q}, \vec{r}$ be three unit vectors such that $\vec{p} \times \vec{q} = \vec{r}$. If $\vec{a}$ is any vector such that $[\vec{a}\ \vec{q}\ \vec{r}] = 1$, $[\vec{a}\ \vec{r}\ \vec{p}] = 2$ and $[\vec{a}\ \vec{p}\ \vec{q}] = 3$ then $\vec{a}$ =

(A) $\vec{p} + 3\vec{q} + \vec{r}$
(B) $3\vec{p} + 2\vec{q} + \vec{r}$
(C) $\vec{p} - 2\vec{q} - 3\vec{r}$
(D) $\vec{p} + 2\vec{q} + 3\vec{r}$

7) Let $P_1 = x + y + z + 1 = 0$, $P_2 = x - y + 2z + 1 = 0$, $P_3 = 3x + y + 4z + 7 = 0$ be three planes. Find the distance of line of intersection of planes $P_1 = 0$ and $P_2 = 0$ from the plane $P_3 = 0$.

(A) $\frac{2}{\sqrt{26}}$
(B) $\frac{4}{\sqrt{26}}$
(C) $\sqrt{\frac{1}{26}}$
(D) $\frac{7}{\sqrt{26}}$

For More Material Join: @JEEAdvanced_2025

8) Let $\vec{v}$ be a unit vector which follows the equation, $\vec{v} \times \vec{b} = \vec{c}$. Also, $|\vec{b}| = 2$ and $|\vec{c}| = \sqrt{3}$ then

(A) $\vec{v} = -\vec{b} + \vec{b} \times \vec{c}$

(B) $\vec{v} = \frac{3}{4}\left(\vec{b} + 2\vec{b} \times \vec{c}\right)$

(C) $\vec{v} = \frac{1}{4}\left(\vec{b} + \vec{b} \times \vec{c}\right)$

(D) $\vec{v} = \frac{\vec{b} \times \vec{c}}{4}$

9) Given $f(x) = \text{sgn}(x^2 - 4|x| + \lambda)$ (where sgn(x) denotes signum fuction of x, $\lambda \in R$)
In the List-I there are values of λ and in the List-II there are number of points where $f(x)$ is non-differentiable.
Match List-I with List-II and select the correct answer using the code given below the list.

List-I		List-II	
(A)	$\lambda = 0$	(I)	0
(B)	$\lambda = 4$	(II)	3
(C)	$\lambda = 5$	(III)	4
(D)	$\lambda = 2$	(IV)	2

(A) (A)-II, (B)-IV, (C)-I, (D)-III

(B) (A)-II, (B)-I, (C)-III, (D)-IV

(C) (A)-I, (B)-II, (C)-III, (D)-IV

(D) (A)-IV, (B)-II, (C)-I, (D)-III

10) Complete solution set of equation $[\cos^{-1}x] + 2[\sin^{-1}x] = 0$ is (where [.] is greatest integer function)

(A) (cos1,sin1)

(B) [–sin1,0) ∪ (cos1,sin1)

(C) [0,1]

(D) [–sin1,cos2] ∪ (cos1,sin1)

11) Let $f(x) = \begin{cases} \frac{\ell n\left(8+x^3\right) - \ell n\left(8-x^3\right)}{x^3}, & \text{if } x \neq 0 \\ k, & \text{if } x = 0 \end{cases}$
If f(x) is continuous at x = 0, then the value of k is

(A) 1

(B) $\frac{1}{2}$

(C) $\frac{1}{4}$

For More Material Join: @JEEAdvanced_2025

(D) $\frac{1}{8}$

12) The volume of the tetrahedron formed by the planes whose equations are $y + z = 0$, $z + x = 0$, $x + y = 0$ and $x + y + z = 2$ is

(A) $\frac{4}{3}$

(B) $\frac{8}{3}$

(C) $\frac{16}{3}$

(D) 32

13) The length of projection of the line segment joining the point (1, 0, –1) and (–1, 2, 2) to the plane $x + 3y - 5z = 6$, is equal to

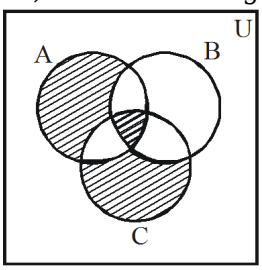
(A) 2

(B) $\sqrt{\frac{271}{53}}$

(C) $\sqrt{\frac{472}{31}}$

(D) $\sqrt{\frac{474}{35}}$

14) The shaded region in venn-diagram can be represented by which of the following?



(A) $(A \cup C) \cap \left(A^C \cup B^C\right) \cup \left(A^C \cup C^C\right) \cup \left(B^C \cup C^C\right)$

(B) $(A \cup C) \cap \left(A^C \cup B^C\right) \cap \left(A^C \cup C^C\right) \cap \left(B^C \cup C^C\right)$

(C) $\left[(A \cup C) \cap \left(A^C \cup B^C\right) \cap \left(A^C \cup C^C\right) \cap \left(B^C \cup C^C\right)\right] \cup (A \cap B \cap C)$

(D) $(A \cup C) \cap \left(A^C \cup B^C\right) \cap \left(A^C \cup C^C\right) \cap \left(B^C \cup C^C\right) \cap (A \cap B \cap C)$

15) **Statement-I:** $\cos^{-1}x = 3\cos^{-1}a$ has a solution for all $a \in [-1, 1]$
and
Statement-II: $\cos^{-1}a$ is defined when $a \in [-1, 1]$

(A) Statement-I is true but Statement-II is false.

(B) Both Statement-I and Statement-II are false.

(C) Both Statement-I and Statement-II are true.

For More Material Join: @JEEAdvanced_2025

(D) Statement-I is false but Statement-II is true.

16) Let R be a relation from A = {2,3,4,5} to B = {3,6,7,10} defined by R = {(a,b) | a divides b, a ∈ A, b ∈ B}, then number of elements in R^{-1} will be

(A) 0
(B) 3
(C) 4
(D) 5

17) Let $f(x) = \left[\frac{x^{2x^4}}{x^{x^2}+3}\right]$, then unit digit of $f(10)$ is (where [.] is greatest integer function)

(A) 2
(B) 4
(C) 3
(D) 1

18) The value of $\lim_{x\to 0} \frac{1-(\cos 2x)^3(\cos 5x)^5(\cos 7x)^7(\sec 4x)^9(\sec 6x)^{11}}{x^2}$ is -

(A) – 46
(B) – 23
(C) – 30
(D) 46

19) Let $f(n) = (2)(4)(6).....(2n)$, $n \in N$ and $g(n) = (1)(3)(5).....(2n-1)$, $n \in N$, then the divisor of $f(100) - g(100)$ is-

(A) 101
(B) 151
(C) 191
(D) 201

20) Let $g(x) = f(x)\sin x$ where $f(x)$ is twice differentiable function on R such that $f'(-\pi) = 1$, then value of $g''(-\pi)$ is equal to -

(A) 1
(B) 2
(C) – 2
(D) 0

SECTION-II

For More Material Join: @JEEAdvanced_2025

1) If value of $\lim_{n\to\infty}\left(2017^n+2018^n\right)^{1/n}$ is L, then the value of $\frac{L}{1009}$ is

2) Given $\vec{a},\vec{b},\vec{c}$ are 3 vectors such that $\vec{b}$ and $\vec{c}$ are unit like parallel vectors and $|\vec{a}|=3$. If $\vec{a}+\lambda\vec{c}=4\vec{b}$, then the sum of all possible value(s) of λ is

3) Let $f(x) = x + 1$, $g^{-1}(x) = x^3 + x + 1$ and $h(x) = g(f(x))$, then $\lim_{x\to1}\frac{h^{-1}(x)-h^{-1}(1)}{x-1}$ is equal to

4) Number of point(s) where $g(x) = |x(2x + 1)(2x - 1)|\cos(\pi x)$ is not differentiable is

5) Absolute value of a for which the lines $L_1:\frac{x-2}{1}=\frac{y-9}{2}=\frac{z-13}{3}$ and $L_2:\frac{x-a}{1}=\frac{y-7}{-2}=\frac{z+2}{3}$ intersect, is

For More Material Join: @JEEAdvanced_2025

ANSWER KEYS

PART-1 : PHYSICS

SECTION-I

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	A	D	A	A	B	B	D	D	A	A	B	A	D	A	D	D	B	B	B	A

SECTION-II

Q.	21	22	23	24	25
A.	2	41	11	27	95

PART-2 : CHEMISTRY

SECTION-I

Q.	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
A.	A	D	B	D	A	D	C	A	B	B	C	A	B	C	A	D	D	B	B	D

SECTION-II

Q.	46	47	48	49	50
A.	5	2	182	50	5

PART-3 : MATHEMATICS

SECTION-I

Q.	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70
A.	D	C	B	A	B	D	B	C	A	D	C	C	D	C	D	D	C	C	D	C

SECTION-II

Q.	71	72	73	74	75
A.	2	8	4	1	3

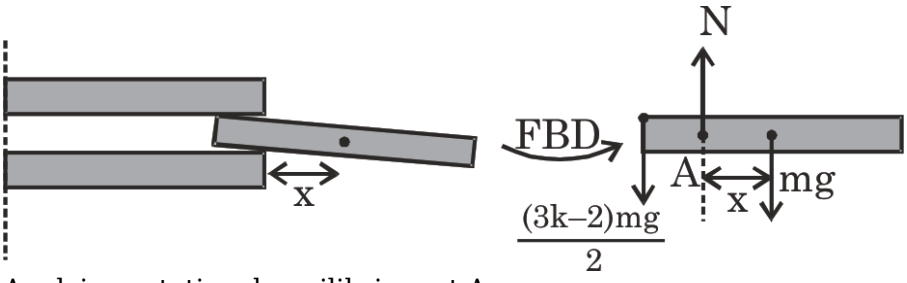
For More Material Join: @JEEAdvanced_2025

SOLUTIONS

PART-1 : PHYSICS

1)

For k^{th} domino



Applying rotational equilibrium at A :-

$$\frac{(3k-2)mg}{2}\left(\frac{\ell}{2}-x\right)=mgx$$

$$\frac{(3k-2)mg}{2}\left(\frac{\ell}{2}-x\right)=\frac{x}{\frac{\ell}{2}-x}$$

$$\frac{\ell}{2}+x=\frac{\ell}{2}\left(\frac{6x-2}{3k}\right)=\left(\frac{3k-1}{3k}\right)\ell$$

Now, $\frac{\ell}{2}+x=\left(1-\frac{1}{3k}\right)\ell$

we can see maximum length of balcony is given by k = n

2)

Mass = ρ. L

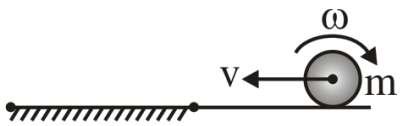
$2\pi R = L$

$$R=\frac{L}{2\pi}$$

$$I=\frac{MR^2}{2}+MR^2=\frac{3}{2}MR^2$$

$$I=\frac{3}{2}.\rho.L.\left[\frac{L}{2\pi}\right]^2$$

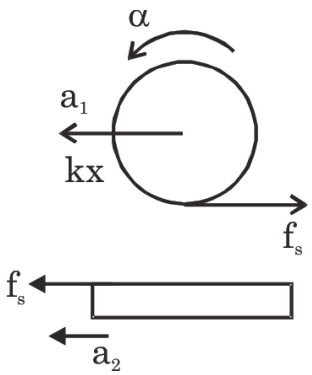
$$I=\frac{3\rho L^3}{8\pi^2}$$



4) A B

Upon return journey the disc rolls with forward slipping. Hence kinetic friction μmg will act on the disc.

For More Material Join: @JEEAdvanced_2025



5)

$kx - f_s = ma_1$ .....(1)

$f_sR =$ .....(2)

$f_s = ma_2$ .....(3)

$a_1 - \alpha r = a_2$ .....(4)

from (2) & (3)

$ma_2R = \frac{mR^2}{2}\alpha$

$a_2 = \frac{\alpha R}{2}$

from equation (4)

$a_1 - 2a_2 = a_2 \Rightarrow a_1 = 3a_2$

6) More i, more brightness because resistance of each bulb is same and i can be found using series & parallel concepts.

7) As secondary battery have P.D. more than potential diffrence across wire AB. Therefore balance point will not be achieved on wire AB.

8)

15 x

□ 100 – □

$\frac{15}{x} = \frac{\ell}{100-\ell}$

x 15

□ – 20 120 – □

$\frac{x}{15} = \frac{\ell-20}{120-\ell}$

9)

(P) For converging lens (convex lens)

$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$\frac{1}{y} + \frac{1}{x} = \frac{1}{d}$

$\frac{1}{y} = \frac{1}{d} - \frac{1}{x}$

at x = 0 y = 0

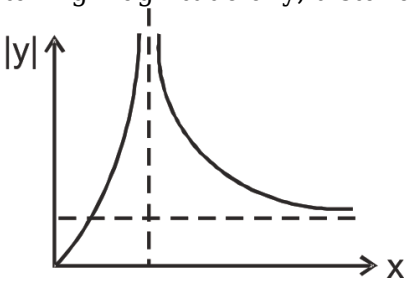
For x = 0 to x = d, y = – ve

so, if x ↑ y ↓ and |y| ↑

At x = d, y = ∞

For More Material Join: @JEEAdvanced_2025

when x > d, y + ve, and
at x = ∞ , y = d
taking magnitude of y, distance graph is shown.



(Q) For converging mirror (concave mirror)

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$

$u = -x,\ f = -\frac{R}{2},\quad v = y$

$$\frac{1}{y}-\frac{1}{x}=-\frac{2}{R}$$

$$\frac{1}{y}=\frac{1}{x}-\frac{2}{R}$$

At x = 0 , y = 0

for $0 < x < \frac{R}{2}$, y = +ve

and as x increases $\frac{1}{y}$ decrease so y ↑ upto $x = \frac{R}{2}$

At $x = \frac{R}{2}$, y = ∞

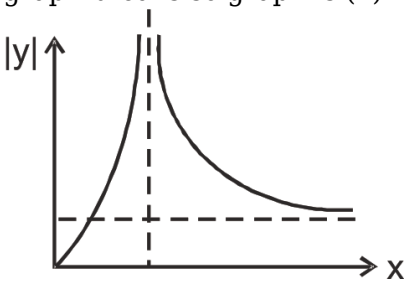
So, graph is (1)

when $x > \frac{R}{2}$ y (–ve)

and as x ↑, 1/y ↓, y ↑ so, |y| ↓

At $x = \infty,\ y = -\frac{R}{2}$

graph breaks so graph is (1)



(R) For diverging Lens (concave lens)

$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$$

u = – x, f = – d v = y

$$\frac{1}{y}+\frac{1}{x}=-\frac{1}{d}$$

$$\frac{1}{y}=-\frac{1}{x}-\frac{1}{d}$$

⇒ y is always –ve

At x = 0, y = 0

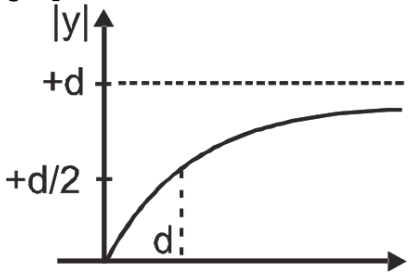
As x ↑, y ↓ so, |y| ↑

For More Material Join: @JEEAdvanced_2025

At x = d, $y = \frac{-d}{2}$

or $x = \infty$, $y = -d$

graph is (2)



(S) For diverging Mirror (convex mirror)

$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

u = – x, $f = +\frac{R}{2}$, v = y

$\frac{1}{y} - \frac{1}{x} = \frac{2}{R}$ $\Rightarrow \frac{1}{y} = \frac{1}{x} + \frac{2}{R}$ $\Rightarrow$ y = +ve

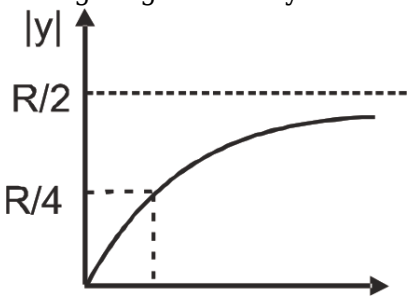
At x = 0, y = 0

$\frac{dy}{dx} = \frac{y^2}{x^2}$

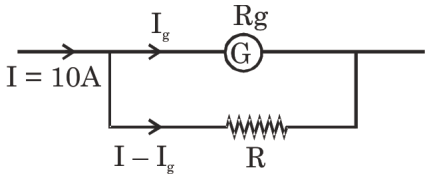
x ↑, y↑

At $x = \frac{R}{2}$, $y = \frac{R}{4}$, At $x = \infty$, $y = \frac{R}{2}$

taking magnitude of y distance graph is



graph is (2)



10)

$(I - I_g)R = I_g R_g$

$R = \frac{I_g R_g}{I - I_g}$

11) As $R \propto \ell^2$ (for stretching)

$\frac{\Delta R}{R}\% = 2\frac{\Delta \ell}{\ell}\%$

$\frac{\Delta R}{R}\% = 2\ (2\%) = 4\%$

For More Material Join: @JEEAdvanced_2025

12)

As $\frac{R_1R_2}{R_1+R_2}=\frac{10}{3}$ and $R_1 + R_2 = 15$

then $R_1R_2 = 50$

Now $R_2-R_1=\sqrt{(R_1+R_2)^2-4R_1R_2}$

$=\sqrt{(15)^2-(50)^2(4)}=5$

Hence $\Rightarrow R_2 = 10\Omega$ and $R_1 = 5\Omega$

$\therefore\frac{\ell_1}{\ell_2}=\frac{5}{10}=\frac{1}{2}$

13) Given, focal length of objective, $f_0 = 30$ cm

Focal length of eye lens, $f_e = 3.0$ cm

Magnifying power, M = ?

Magnifying power of the Galilean telescope,

$M_D=\frac{f_0}{f_e}\left(1-\frac{f_e}{D}\right)=\frac{30}{3}\left(1-\frac{3}{25}\right)$

$=10\times\frac{22}{25}=8.8$ cm [D = 25 cm]

14) Net magnification of compound microscope

$m = m_0 \cdot m_e$

$m_e \rightarrow$ magnification due to eye piece

$m_0 \rightarrow$ magnification due to objective

When final image is at least vision

$m_e=\left(1+\frac{D}{f_e}\right)=\left(1+\frac{25}{5}\right)=6$

$m = m_0 \cdot m_e$

$m_0=\frac{30}{6}=5$

$m_0 = 5$

16) For small angled prism $\delta=(\mu-1)A\Rightarrow\delta=\mu A-A$ from graph $A = 4^0$, slope = A =4

17) $\frac{1}{F_e}=\frac{1}{f_m}-\frac{2}{f_1}\times 4$

18) The image formed first by refraction in the convex lens is real and inverted and of the same size as the object and is obtained at the distance $2f_1$ from the lens. This real image is then reflected once in the converging mirror. The object distance for this reflection is given by $u = (d - 2f_1)$. Now this reflection should yield a real and inverted image which again should have the same size as the object.

Obviously, the object must be at the centre of curvature of this converging mirror. Hence we must have u = radius of curvature of concave mirror i.e., $u = d - 2f_1 = 2f_2 \Rightarrow d = 2f_1 + 2f_2 = 2(f_1 + f_2)$

For More Material Join: @JEEAdvanced_2025

19)

M.S. reading 35 mm
X + 4V = nM
X = nM – $4\left(\frac{5}{8}M\right)$
X < 1 mm for n = 3 only
So, $X = 3 - \frac{5}{2} = \frac{1}{2}mm$
 reading = 35 + 0.5 mm
= 3.55 cm

20) $\frac{3Rx}{3R+x} + 2R + 2R = x$
⇒ x = 6R

21) 10 VSD = 8 MSD
$1VSD = \frac{8 \times 1mm}{10}$
L.C. = 1 MSD – 1 VSD = 0.2 mm

22) $u = \frac{h}{h'} = \frac{5.25}{5.00}$
Least count = $\frac{1}{20}cm - \frac{49}{50} \cdot \frac{1}{20}cm$
$= \frac{1}{50} \times \frac{1}{20}cm = 0.01\ mm$
ln u = ln h – ln h′
$\frac{du}{u} = \frac{dh}{h} - \frac{dh'}{h'}$
$du = \left[\frac{0.01}{5.25} + \frac{0.01}{5.00}\right] \frac{5.25}{5.00}$
$= \frac{41}{10} \times 10^{-3}$

23)

Diameter D = 0.050 cm, ΔD = 0.001 cm
Length L = 110.0 cm
ΔL = 0.1 cm
Weight W = 50 N
Extension = 0.125 cm = $\Delta\ell$ = 0.001 cm
$Y = \frac{\frac{W}{A}}{\frac{\ell}{L}} \Rightarrow Y = \frac{4WL}{\pi D^2 \ell}$
$Y = \frac{4 \times 50 \times 1.1}{\pi \times 25 \times 10^{-8} \times 125 \times 10^{-5}} \frac{N}{m^2}$
⇒ $Y = 0.02241 \times 10^{13}\ N/m^2$
$Y = 2.241 \times 10^{11}\ N/m^2$

For More Material Join: @JEEAdvanced_2025

$$\Rightarrow \frac{\Delta Y}{Y} = \frac{\Delta L}{L} + \frac{2\Delta D}{D} + \frac{\Delta \ell}{\ell}$$

$$\Rightarrow \frac{\Delta Y}{Y} = \frac{0.1}{110} + 2 \times \frac{0.001}{0.050} + \frac{0.001}{0.125}$$

$$\Rightarrow \frac{\Delta Y}{Y} = 0.04891$$

$\Rightarrow \Delta Y = Y \times 0.04891$

$\Rightarrow \Delta Y = 2.241 \times 10^{11} \times 0.04891$

$\Rightarrow \Delta Y = 0.1096 \times 10^{11}$ N/m^2

$\Rightarrow \Delta Y = 1.096 \times 10^{10}$ N/m^2

Least significant digit in this operation is two

$\Delta Y = 1.1 \times 10^{10}$ N/m^2 = 11×10^9 *N/m*^{2}

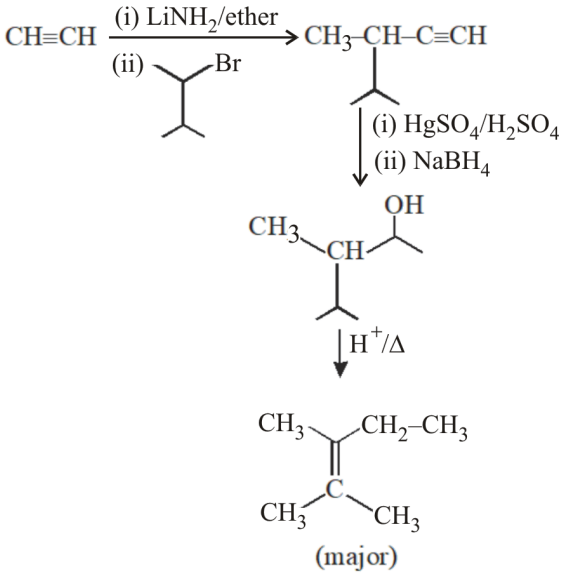
24) $\frac{\in}{r} = 6A, \quad \frac{\in}{r+4.5} = 3A$

$\frac{r+4.5}{r} = 2 \Rightarrow r = 4.5 \Rightarrow \in = 27$

25) Deviation of the red light is $\delta_r = (\mu_r - 1)$ A and deviation of the violet light is $\delta_v = (\mu_v - 1)A$.
he dispersion $\theta = \delta_v - \delta_r = (\mu_v - \mu_r)A$
$= (1.632 - 1.613) \times 5°$
$\theta = 0.095°$ Hence Answer is 1000 θ = 95 degree

PART-2 : CHEMISTRY

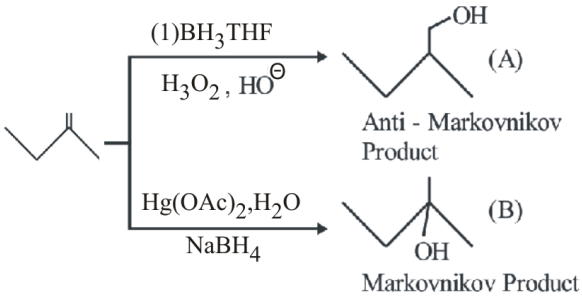
26) Fact based.



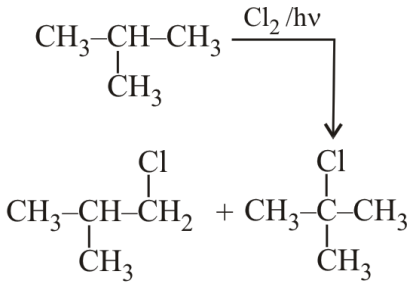
28) Now :- (i) $HgSO_4$/dil.H_2SO_4
(ii) $NaBH_4$
is convert triple bond into ketone and formed ketone is reduced by $NaBH_4$ and convert into Alcohol.

For More Material Join: @JEEAdvanced_2025

30)



32)



(Major)

33)

	KNO_3	$BaCl_2$	Sucrose	$CuSO_4$
C =	0.05M	0.04M	0.140M	0.075M
i =	2	3	1	2
C_{net} =	(i × C)			
C_{net}	0.1	0.12	0.14	0.15

$\Delta T_f \propto C_{net}$

If C_{net} increases, ΔT_f also increases but freezing point decreases.

34) $K = Ae^{-\Delta H/RT}$

$k = Ae^{-E_a/RT}$

35)

$LaCl_3(s) + H_2O(g) \rightleftharpoons LaClO(s) + 2HCl(g)$ - heat already at equilibrium, more water vapour is added without altering T or V of the system. When equilibrium re-established, the pressure of water vapour is doubled. The pressure of HCl present in the system increases by a factor of $2^{1/2}$.

The equilibrium constant expression is

$$K = \frac{[HCl]^2}{[H_2O]}$$

Before addition of water vapour,

$$K = \frac{[HCl]_i^2}{[H_2O]_i} \quad ....(1)$$

After addition of water vapour,

For More Material Join: @JEEAdvanced_2025

$$K = \frac{[HCl]_f^2}{[H_2O]_f} \text{ .....(2)}$$

But (1) = (2)

$$\frac{[HCl]_i^2}{[H_2O]_i} = \frac{[HCl]_f^2}{[H_2O]_f} \text{ .....(3)}$$

When equilibrium re-established, the pressure of water vapour is doubled.

$[H_2O]_f = 2[H_2O]_i$ .....(4)

Substitute (4) in (3)

$$\frac{[HCl]_i^2}{[H_2O]_i} = \frac{[HCl]_f^2}{(2[H_2O]_i)}$$

$$[HCl]_i^2 = [HCl]_f^2 \times \frac{1}{2}$$

$$[HCl]_i = [HCl]_f \times \sqrt{\frac{1}{2}}$$

$$[HCl]_f = \sqrt{2} \times [HCl]_i$$

36) $[Ag^+][I^-] = K_{SP}[AgI]$

$$[Ag^+] = \frac{10^{-17}}{10^{-2}} = 10^{-15}$$

37) First dissociation : $X(OH)_3 \rightarrow X(OH)_2^+ + OH^-$

Second dissociation : $X(OH)_2^+ \rightarrow X(OH)^{2+} + OH^-$

Total $[OH^-] = 4 \times 10^{-3} + 2 \times 10^{-3} = 6 \times 10^{-3}$

$p_{OH} = 3 - \log 6 = 2.22$

$p_H = 11.78$

38) $$Y_A = \frac{X_A p_A^o}{X_A(p_A^o - p_B^o) + p_B^o}$$

$$\Rightarrow \frac{1}{Y_A} = \frac{(p_A^o - p_B^o)}{p_A^o} + \left(\frac{p_B^o}{p_A^o}\right) \times \frac{1}{X_A}$$

$$\Rightarrow \frac{1}{X_A} = \frac{(p_B^o - p_A^o)}{p_B^o} + \left(\frac{p_A^o}{p_B^o}\right) \times \frac{1}{Y_A}$$

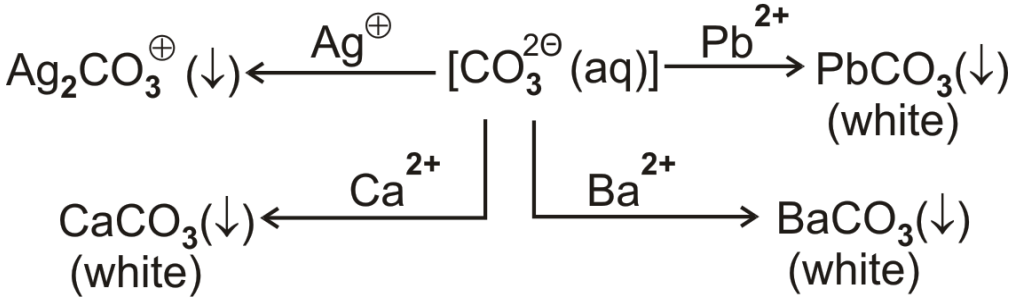
$$\text{Slope} = \frac{p_A^o}{p_B^o} \text{ intercept } \frac{p_B^o - p_A^o}{p_B^o}$$

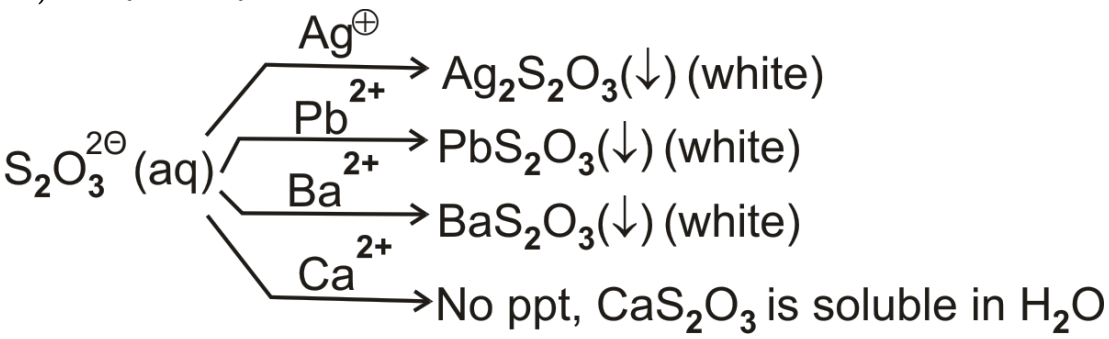
39) NO_3^- ion + $FeSO_4$ + Conc. $H_2SO_4 \rightarrow$ Brown ring complex

40) "MnO_2" has less oxidising action F_2. Hence F^- can't be oxidised by MnO_2.

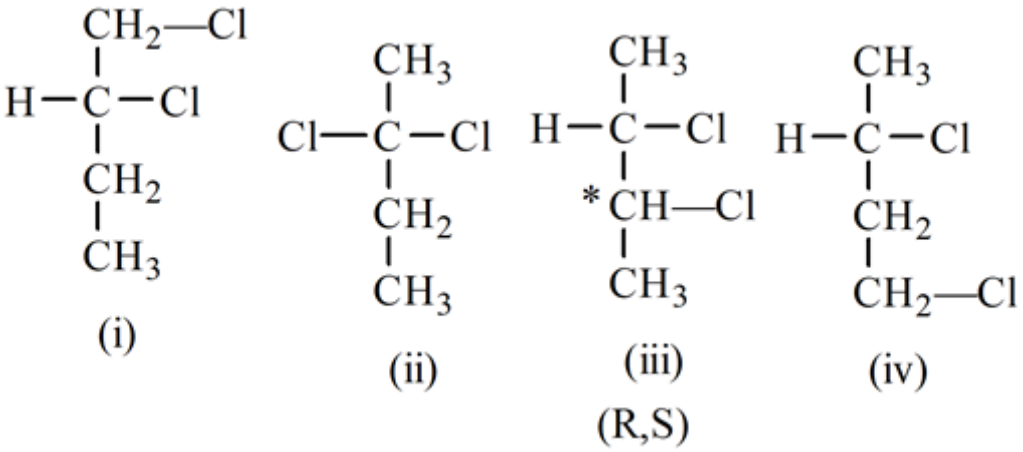
For More Material Join: @JEEAdvanced_2025

41)





46)



47)

$Y_A = 0.5 \Rightarrow Y_B = 0.5$
$P_A = P_B = 0.4$ atm
$P_A = P_A^0 X_A$
$P_A^0 = 2$ atm

48)

$A + B \rightleftharpoons C + D : K_{eq} = 100$
1M 1M 1M 1M
First check direction of reversible reaction.
Since $Q_C = \frac{[C][D]}{[A][B]} = 1 < K_{eq.}$ ⇒ reaction will move in forward direction to attain equilibrium state.
$\Rightarrow$ $A + B \rightleftharpoons C + D : K_{eq} = 100$
to 1 1 1 1
$t_{eq.}$ 1–x 1–x 1+x 1+x

For More Material Join: @JEEAdvanced_2025

Now : $K_{eq} = 100 = \frac{(1+x)(1+x)}{(1-x)(1-x)}$

$\Rightarrow \boxed{100 = \left(\frac{1+x}{1-x}\right)^2}$

(i) $10 = \left(\frac{1+x}{1-x}\right)$

$\Rightarrow 10 - 10x = 1+x$

$\Rightarrow 11x = 9$

$\Rightarrow \boxed{x = \frac{9}{11}}$

(ii) $-10 = \frac{1+x}{1-x}$

$\Rightarrow -10 + 10x = 1 + x$

$\Rightarrow -9x = -11$

$\Rightarrow \boxed{x = \frac{11}{9}}$

→ 'x' cannot be more than one, therefore not valid.

therefore equation concretion of (D) = 1 + x

$= 1 + \frac{9}{11} = \frac{20}{11}$

$= 1.8181 = 181.81 \times 10^{-2}$

$= 182 \times 10^{-2}$

49) A_2X ; $K_{sp} = 4s^3 \Rightarrow S = \left(\frac{K_{ap}}{4}\right)^{1/3}$

$S_{(A_2X)} = \left(\frac{4 \times 10^{-12}}{4}\right)^{1/3} = 10^{-4}M$

MX; $K_{sp} = s^2$, $s = \sqrt{K_{sp}}$

$S_{(MX)} = \sqrt{4 \times 10^{-12}} = 2 \times 10^{-6}$ M

$\frac{S_{A_2X}}{S_{MX}} = \frac{10^{-4}}{2 \times 10^{-6}} = 50$

50) PbS, CuS Bi_2S_3, CoS, NiS → Black

$Fe(OH)_3$ → Brown

Sb_2S_5 → Orange

$Cr(OH)_3$ → Green

MnS, → Pink

$BaCO_3$, $SrCO_3$ → White

PART-3 : MATHEMATICS

51) Diagonals are $6\vec{a} - \vec{b}$ and $4\vec{a} + 5\vec{b}$

For More Material Join: @JEEAdvanced_2025

Lengths are $\sqrt{36a^2 + b^2 - 12ab\cos\frac{\pi}{4}}$
and $\sqrt{16a^2 + 25b^2 + 40ab\cos\frac{\pi}{4}}$
 length of smallest diagonal is 15

52) $f(1) = 0$
$f(1 + h) = \lim_{h\to 0}(|(1 + h)| - 1)\operatorname{sgn} h = 0$
$f(1 - h) = 0$
at x = 0 both (|x| – 1) & sgn(x – 1) are continuous and their product is also continuous.

53)

$f(x - 1) + f(x + 3) = f(x + 1)$ .....(i)
Replacing x by x + 2
$f(x + 1) + f(x + 5) = f(x + 3)$ .....(ii)
from (i) and (ii)
$f(x - 1) + f(x + 5) = 0$
Replacing x by x + 1
$\Rightarrow f(x) + f(x + 6) = 0$
$\Rightarrow f(x) = -f(x + 6)$ .....(v)
Replacing x by x + 6
$f(x + 6) = -f(x + 12)$
$\Rightarrow f(x) = f(x + 12)$
hence f(x) is periodic function with period 12
Now $\sum_{r=0}^{90} f(5 + 12r) = f(5) + f(17) + ... + f(1085)$ = 91
hence **A** is not correct but **R** is correct.

54) Let $\vec{P} = \lambda(\hat{i} + 2\hat{j} + 3\hat{k}) \times (-2\hat{i} + 4\hat{j} + \hat{k})$
$= -10\lambda\hat{i} - 7\lambda\hat{j} + 8\lambda\hat{k}$
given $\vec{P}.\left(\hat{i} - 2\hat{j} + \hat{k}\right) = -6$
$\Rightarrow -10\lambda + 14\lambda + 8\lambda = -6$
$\Rightarrow \lambda = \frac{-1}{2}$
 $\vec{P} = 5\hat{i} + \frac{7}{2}\hat{j} - 4\hat{k}$
$\Rightarrow \vec{P}.\hat{i} + \vec{P}\hat{k} = 5 - 4 = 1$

55) $n(\bar{A} \cap \bar{B}) = n\left(\overline{A \cup B}\right)$
= n(u) – n(A ∪ B)
= 600 – (n(A) + n(B) – n(A ∩ B))
= 600 – 100 – 200 + 50
= 350

For More Material Join: @JEEAdvanced_2025

56) Given that $\vec{p} \times \vec{q} = \vec{r}$

$\Rightarrow (\vec{p} \times \vec{q}) \cdot \vec{r} = |\vec{r}|^2$

$\Rightarrow [\vec{p}\,\vec{q}\,\vec{r}] = 1$

Let $\vec{a} = x\vec{p} + y\vec{q} + z\vec{r}$ .....(1)

Take dot product in equation (1) both sides with $\vec{p} \times \vec{q}$, $\vec{q} \times \vec{r}$ and $\vec{r} \times \vec{p}$ we get x = 1, y = 2, z = 3.

57) Let direction ratios of line are a, b, c

Now a + b + c = 0

a – b + 2c = 0

$\frac{a}{3} = \frac{b}{-1} = \frac{c}{-2}$

So direction ratios of line are (3, –1, –2)

x + y + z + 1 = 0 .....(1)

x – y + 2z + 1 = 0 .....(2)

Put z = 0, we get x = 1, y = 0

Equation of line of intersection of $P_1 = 0$, $P_2 = 0$ is parallel to the plane $P_3 = 0$.

So distance of point (–1, 0, 0) from $P_3 = 0$ is $\left|\frac{-3+7}{\sqrt{9+1+16}}\right| = \frac{4}{\sqrt{26}}$

58) $\vec{b} \times (\vec{v} \times \vec{b}) = \vec{b} \times \vec{c}$

$4\vec{v} - (\vec{b}.\vec{v})\,\vec{b} = \vec{b} \times \vec{c}$

$|\vec{v} \times \vec{b}| = |\vec{c}|$

$\Rightarrow |\vec{v}|\,|\vec{b}| \sin\theta = |\vec{c}|$

$2|\vec{v}| \sin\theta = \sqrt{3}$

$\sin\theta = \frac{\sqrt{3}}{2}$

$\cos\theta = \frac{1}{2}$

$\Rightarrow 4\vec{v} - \left(2 \times 1 \times \frac{1}{2}\right)\vec{b} = \vec{b} \times \vec{c}$

$\vec{v} = \frac{1}{4}(\vec{b} + \vec{b} \times \vec{c})$

59)

For λ = 0, $f(x) = \text{sgn}(x^2 - |x|)$

$f(x)$ is non diff. at x = 0,4,–4

for λ = 4, $f(x) = \text{sgn}(((x)-2)^2)$

 Non diff at x = 12

for λ = 5, $f(x) = \text{sgn}\left(\underbrace{x^2 - 4|x| + 5}_{D<0}\right)$

 differentiable ∀ x ∈ R

for λ = 2, $f(x) = \text{sgn}(x^2 - 4|x| + 2)$

For More Material Join: @JEEAdvanced_2025

 Non diff. at 4 points.

60) $[\cos^{-1}x] + 2[\sin^{-1}x] = 0$
only if $[\cos^{-1}x] = [\sin^{-1}x] = 0$
or $[\cos^{-1}x] = 2, [\sin^{-1}x] = -1$
Case-I : $[\cos^{-1}x] = 0 \Rightarrow x \in (\cos 1, 1]$ ....(1)
$[\sin^{-1}x] = 0 \Rightarrow x \in [0, \sin 1)$ ....(2)
$\Rightarrow$ (1) $\cap$ (2), $x \in (\cos 1, \sin 1)$ ....(3)
Case-II : $[\cos^{-1}x] = 2 \Rightarrow x \in (\cos 3, \cos 2]$ ....(4)
$[\sin^{-1}x] = -1 \Rightarrow x \in [-\sin 1, 0)$ ....(5)
(3) $\cup$ (5) $\Rightarrow [-\sin 1, \cos 2] \cup (\cos 1, \sin 1)$

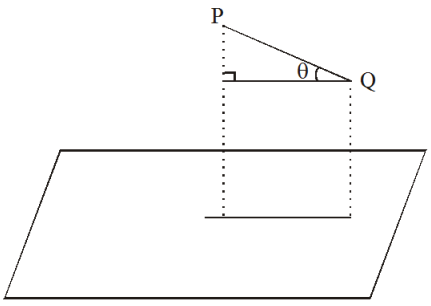
61) $$k = \lim_{x\to 0} \frac{\ell n\left(\frac{8+x^3}{8-x^3}\right)}{x^3} = \lim_{x\to 0} \frac{2x^3}{(8-x^3)\,x^3} = \frac{1}{4}$$

62) The planes are
$y + z = 0$ ...(1)
$z + x = 0$ ...(2)
$x + y = 0$ ...(3)
$x + y + z = 2$ ...(4)
on solving, vertices are (0, 0, 0), (2, 2, – 2), (2, – 2, 2) and (–2, 2, 2).

Required volume $= \frac{1}{6}\begin{vmatrix} 2 & 2 & -2 \\ 2 & -2 & 2 \\ -2 & 2 & 2 \end{vmatrix}$

$= \frac{8}{6}\begin{vmatrix} 1 & 1 & -1 \\ 1 & -1 & 1 \\ -1 & 1 & 1 \end{vmatrix} = \frac{16}{3}$

63) Equation of line $\frac{x-1}{-2} = \frac{y}{2} = \frac{z+1}{3}$
Let P(1, 0, –1), Q (–1, 2, 2)
$\overrightarrow{PQ} = -2\hat{i} + 2\hat{j} + 3\hat{k}$
$\left|\overrightarrow{PQ}\right| = \sqrt{4+4+9} = \sqrt{17}$
angle between line & plane
$\sin\theta = \left|\frac{-2.1 + 2.3 - 5.3}{\sqrt{1+9+25}\sqrt{17}}\right| = \frac{11}{\sqrt{35}\sqrt{17}} = \frac{11}{\sqrt{595}}$
$\cos\theta = \frac{\sqrt{474}}{\sqrt{595}}$
Projection $= \left|\overrightarrow{PQ}\right| \cos\theta = \sqrt{17}.\frac{\sqrt{474}}{\sqrt{35}\sqrt{17}} = \sqrt{\frac{474}{35}}$



64) Obviously option (3) is correct.

For More Material Join: @JEEAdvanced_2025

65) $0 \le 3\cos^{-1}a \le \pi$

$$0 \leqslant \cos^{-1}a \leqslant \frac{\pi}{3} \Rightarrow \frac{1}{2} \leqslant a \leqslant 1$$

66) R = {(2,6), (2,10), (3,3), (3,6),(5,10)}

 5 elements in R^{-1}

67) $f(10) = \left[\frac{10^{2\times10^4}}{10^{100}+3}\right]$; let $10^{100} = t$

$$\Rightarrow f(10) = \left[\frac{t^{200}}{t+3}\right] = \left[\left(\frac{(t^2)^{100}-(3^2)^{100}}{t+3}\right) + \left(\frac{9^{100}}{t+3}\right)\right]$$

integer ↓ fractions ↓

$\Rightarrow f(10) = \frac{10^{20000}-81^{50}}{10^{100}+3}$ $\Rightarrow$ unit digit is 3.

 x i.e. 201 is a divisor

68) $$\lim_{x\to0} \frac{(\cos 6x)^{11}(\cos 4x)^9 - (\cos 2x)^3(\cos 5x)^5(\cos 7x)^7}{x^2}$$

$$\lim_{x\to0} \frac{(1-18x^2)^{11}(1-8x^2)^9 - (1-2x^2)^3\left(1-\frac{25x^2}{2}\right)^5\left(1-\frac{49x^2}{2}\right)^7}{x^2}$$

$\Rightarrow$ L = – 30

69) f(100) – g(100)

= (2)(4)(6) ..... (200) – (1)(3)(5).....(199)

Let x = 201

= (x – 1)(x – 2)....(x – 199) – (1)(3) ..... (199)

as constant term cancelled

70) g(x) = f(x) sinx

g'(x) = f(x)cosx + f'(x) sinx

g"(x) = f'(x)cosx – sinxf(x) + f'(x) cosx + sinxf"(x)

g"(–π) = 2f'(–π)cosπ = –2

71) $$L = 2018\left(\lim_{n\to\infty}\left(\frac{2017}{2018}\right)^n + 1\right) = 2018$$

72) $|\vec{a}| = |4\vec{b} - \lambda\vec{c}|$

$|\vec{a}|^2 = |4\vec{b} - \lambda\vec{c}|^2$

$\Rightarrow 9 = 16\vec{b}^2 + \lambda^2\vec{c}^2 - 8\lambda\vec{b}.\vec{c}$

For More Material Join: @JEEAdvanced_2025

$9 = 16 + \lambda^2 - 8\lambda$

$\Rightarrow \quad \lambda^2 - 8\lambda + 7 = 0$

Sum of all possible values of $\lambda = 8$

73) Required limit is $\left.\frac{d}{dx}\left(h^{-1}(x)\right)\right|_{x=1}$

$\square$ $f^{-1}(x) = x - 1$; $g^{-1}(x) = x^3 + x + 1$

Also $h(x) = g(f(x))$

$\Rightarrow h^{-1}(x) = f^{-1}(g^{-1}(x)) = x^3 + x$

$\Rightarrow \frac{d}{dx}\left(h^{-1}(x)\right) = 3x^2 + 1$

74) g(x) is not differentiable at x = 0 only

75) $L_1 : \frac{x-2}{1} = \frac{y-9}{2} = \frac{z-13}{3} = \lambda$

$\Rightarrow$ $x = 2 + \lambda$, $y = 2\lambda + 9$, $z = 3\lambda + 13$

This lies on L_2

$\square$ $\underbrace{\frac{2+\lambda-a}{1}}_{p} = \underbrace{\frac{2\lambda+9-7}{-2}}_{q} = \underbrace{\frac{3\lambda+13+2}{3}}_{r}$

by q and r : $\lambda = -3$

put in q and solve p and q, $a = -3$

For More Material Join: @JEEAdvanced_2025